

llmXive Automated Review of GENEb: Why Genomic Models Are Hard to Compare

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a comprehensive benchmark (GENEB) and makes several strong claims about the limitations of current genomic model comparisons. The core findings regarding the disconnect between model scale and performance, and the importance of pretraining data alignment, are generally well-supported by the provided statistical summaries and tables.

However, there are specific discrepancies between the textual claims and the data presented in the tables that require correction:

1. **Contradictory Data in Transfer Learning:** In the “Transfer Learning Analysis” section, the text states that human-only pretraining shows an advantage for “Enhancers (+0.010)”. However, Table `tab:transfer_human_multi` explicitly lists the Δ MCC for Enhancers as -0.001 , indicating a slight advantage for multi-species pretraining (or parity), not human-only. This is a direct factual error in the prose that contradicts the provided data table. The text must be corrected to reflect the table’s value.
2. **Overstatement of Statistical Significance:** The text claims that “species classification” shows “no significant scaling” with a p-value of 0.056. While this is technically not significant at the standard $\alpha = 0.05$ threshold, describing it as definitively “no significant scaling” ignores the marginal nature of the result. Given the p-value is very close to 0.05, a more precise phrasing such as “marginally non-significant” would be more accurate and scientifically rigorous.
3. **Generalization of Controlled Pair Results:** The text frequently presents results from specific controlled pairs (e.g., Omni-DNA-1B vs. eccDNAMamba) as indicative of broader architectural trends. While the paper mentions these are controlled comparisons, the prose sometimes implies these are universal properties of the architectures (e.g., “Transformer models show substantial advantages”). It is crucial to ensure the reader understands these are specific findings from the 30 controlled pairs and not necessarily a universal law, especially given the variability seen in other pairs (e.g., GenomeOcean-500M vs. GENA-LM showing a negligible difference).
4. **Traceability of Specific Scores:** In the “Practitioner Recommendations” section, specific numerical claims for 10-shot performance (e.g., MutBERT scoring 0.300 on Histone modifications) are stated without a direct citation to a table or figure in the text body. Since the figures are not visible in the text-only review, these claims are difficult to verify. The text should explicitly reference the specific table or figure where these 10-shot breakdowns are detailed.

Addressing these points will ensure the factual claims in the manuscript are perfectly aligned with the underlying data and statistical evidence.

Action items.

- **[writing]** In ‘Architecture Comparison’, the claim that Omni-DNA-1B exceeds eccDNAMamba by +0.149 is from a specific controlled pair. Ensure text clarifies this is not a universal architecture advantage, as other pairs show negligible differences.

- **[science]** In ‘Transfer Learning Analysis’, the text claims human-only pretraining has an advantage for Enhancers (+0.010), but Table `tab:transfer_human_multi` shows Delta = -0.001 (multi-species advantage). Correct the text to match the table data.
- **[writing]** In ‘Practitioner Recommendations’, the specific 10-shot score for MutBERT on Histone modifications (0.300) is not cross-referenced to a table or figure in the text. Add a citation to the specific data source for verification.
- **[writing]** The claim that species classification shows ‘no significant scaling’ ($p=0.056$) is technically correct at $\alpha=0.05$ but overstates certainty. Rephrase as ‘marginally non-significant’ to accurately reflect the p-value proximity to the threshold.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure visually contradicts its caption by depicting a dense, connected network rather than a sparse, disconnected one. Additionally, the central decorative elements obscure the data, and the high density of overlapping edges makes the graph difficult to interpret.

Figure 2

Figure 2 presents a clear hierarchical taxonomy of genomic foundation models with legible text and logical grouping, but the caption lacks sufficient detail to explain the color-coding and architectural distinctions implied by the diagram.

Figure 3

The figure is visually clear and effectively communicates the efficiency frontier, but the caption contains a factual contradiction regarding what marker size encodes (stating MCC while the legend indicates parameter count) and ends with a truncated sentence.

Figure 4

The heatmap effectively conveys task-level heterogeneity but suffers from poor x-axis label readability and ambiguous y-axis row alignment. The claimed sorting by macro-average MCC is not visually evident in the model ordering.

Figure 5

The figure effectively visualizes category-specific model strengths using radar plots, but the caption is truncated. Additionally, the inclusion of the known outlier ‘Evo-1-131k’ in the lowest rank subplot without specific annotation may be misleading given the context provided in other figure captions.

Figure 6

The figure effectively visualizes performance degradation trends across shot regimes, but the top band’s percentage values lack clear unit definitions, and the crowded x-axis labels compromise model identification.

Figure 7

Figure 7 effectively visualizes the relationship between task variance and pretraining data composition. However, the x-axis units in panel (B) are not explicitly labeled on the axis itself, and the color coding for top/bottom placements lacks a dedicated legend.

Figure 8

The figure is a clear and well-organized taxonomy diagram with no apparent issues in its visual presentation or labeling.

Figure 9

The figure effectively visualizes the robustness of rankings, but the caption contains unrendered LaTeX placeholders and incorrectly describes the sorting order of the models compared to the visual plot.

Figure 10

Figure 10 is a collection of boxplots comparing model performance across various genomic tasks, but it is rendered unusable by a placeholder caption and a lack of clear panel labeling or axis definitions.

Figure 11

The figure effectively visualizes performance distributions across task categories, but the single-task categories (Coding/Non-cod., Chromatin Acc.) display points rather than the boxplots described in the caption, and one data point appears missing in the Coding/Non-cod. subplot.

Figure 12

The figure effectively displays top-10 model performance across categories, but the x-axis labels are cluttered and overlapping, and the y-axis lacks a direct label on the subplots.

Action items.

- **[science]** Figure 1: The caption claims the graph is ‘sparse’ and ‘disconnected,’ but the rendered image shows a dense, fully connected web of edges with no visible gaps, directly contradicting the textual claim.
- **[writing]** Figure 1: The central question mark and circle are purely decorative and obscure the underlying network structure without adding informational value or being defined in the caption.
- **[writing]** Figure 1: The graph is extremely cluttered with overlapping edges and labels (e.g., ‘DNABERT-2’, ‘DNABERT-S’, ‘Genomics-FM’), making it difficult to trace specific connections or verify the ‘fragmented’ claim visually.
- **[writing]** Figure 2: The caption ‘Architectural taxonomy of genomic foundation models’ is too brief for a complex taxonomy diagram; it should define the color-coding scheme (orange, green, blue) used to group models and clarify the distinction between ‘Transformers’ and ‘Efficient Alternatives’ branches.
- **[science]** Figure 2: The diagram lists specific model names (e.g., ‘mutBERT’, ‘Genomics-FM’) but provides no visual or textual indication of their architectural differences beyond the broad ‘Encoder-only’ or ‘Decoder-only’ labels, limiting the figure’s utility for understanding true architectural diversity.
- **[science]** Figure 3: The caption states ‘Marker size and color both encode macro-MCC’, but the legend explicitly defines marker size as encoding ‘Model Size’ (10^7 , 10^8 , 10^9 params). This contradicts the caption and creates confusion, as size is the primary variable on the x-axis.
- **[writing]** Figure 3: The caption text is truncated at the end (‘...offset su’), cutting off the final sentence.
- **[writing]** Figure 4: The x-axis labels (model names) are rotated 45 degrees and overlap significantly, making many names illegible (e.g., ‘GENERA...-3B’, ‘NT-v2-50M-3mer-MS’). Consider horizontal spacing or font size adjustment.
- **[writing]** Figure 4: The y-axis labels (task groups) are not aligned with the heatmap rows, creating ambiguity about which row corresponds to which task (e.g., ‘Chromatin Acc.’ vs ‘Coding/Non-cod.’). Add grid lines or explicit row markers.
- **[science]** Figure 4: The caption states models are ‘sorted by overall full-shot macro-average MCC’, but the x-axis order does not reflect this sorting (e.g., ‘GENERA...-3B’ appears first despite likely lower aggregate performance than later models). Verify sorting logic.
- **[writing]** Figure 5: The caption is truncated mid-sentence at the end (‘These profiles mo’), suggesting missing text.
- **[science]** Figure 5: The legend for the ‘Rank 36-40’ subplot includes ‘Evo-1-131k’, but the caption explicitly identifies this as a ‘prokaryotic-only outlier’ (referenced in Figure 3’s

caption), yet it is included in the general ranking without specific exclusion or annotation in the plot itself, potentially confusing the comparison of general genomic models.

- **[writing]** Figure 5: The legend for ‘Rank 36-40’ lists ‘JanusDNA-72-wo’ and ‘JanusDNA-72-w’ which appear to be very similar model names; ensure these are distinct and correctly labeled.
- **[science]** Figure 6: The top band displays percentage values (e.g., 65%, 63%) but lacks a unit label or explicit definition in the caption clarifying if these represent relative performance retention or absolute percentage point drops, creating ambiguity in interpreting the ‘relative performance drop’ claim.
- **[writing]** Figure 6: The x-axis labels for the 40 models are densely packed and rotated, causing significant overlap and illegibility for many model names (e.g., ‘GENA-LM-Large-T2T’, ‘NT-v2-50M-MS’), hindering the ability to identify specific data points.
- **[writing]** Figure 7: The x-axis label in panel (B) (‘Bottom-3 placements’ / ‘Top-3 placements’) is ambiguous regarding the unit of measurement (e.g., raw count vs. percentage), which is only clarified by the caption’s mention of ‘n = 39 each’.
- **[writing]** Figure 7: Panel (B) uses a diverging bar chart format but lacks a legend explicitly mapping the bar colors (orange vs. green) to the ‘bottom-3’ and ‘top-3’ categories, relying on the axis labels for interpretation.
- **[writing]** Figure 8: The caption is identical to Figure 2’s caption (‘Architectural taxonomy of genomic foundation models’) and the image filename is ‘[fm_models.png]’, suggesting this is a duplicate figure rather than a distinct entry.
- **[writing]** Figure 9: The caption contains unrendered LaTeX placeholders (e.g., ‘= $MCC_{macro} - MCC_{micro}$ ’, ‘= 0.988’, ‘|| = 0.009’) instead of the variable names (likely Δ) or values, making the text difficult to read.
- **[writing]** Figure 9: The caption states models are sorted from ‘largest negative shift to largest positive shift’, but the plot displays them from largest positive shift (top) to largest negative shift (bottom), contradicting the description.
- **[fatal]** Figure 10: The caption is a placeholder (‘Enter Caption’) and does not describe the figure content, violating the requirement for a descriptive caption.
- **[science]** Figure 10: The figure displays 13 distinct subplots of model performance across task categories, but lacks a unifying title or panel labels (e.g., A, B, C) to identify them, making it difficult to reference specific categories in the text.
- **[science]** Figure 10: The y-axis lists numerous model names (e.g., ‘GENERator-Eukaryote-3B’, ‘Omni-DNA-1B’) that are not defined or explained in the placeholder caption, leaving the reader to guess the context of the comparison.
- **[science]** Figure 11: The ‘Coding/Non-cod.’ and ‘Chromatin Acc.’ subplots display single dots instead of boxplots, contradicting the caption’s claim that ‘boxes denote the interquartile range’ and ‘central lines indicate medians’ for these categories.

- **[writing]** Figure 11: The ‘Coding/Non-cod.’ subplot lists 15 models but only displays 14 data points, indicating a missing value or plotting error.
- **[writing]** Figure 12: The x-axis labels (model names) are rotated 45 degrees and overlap significantly, making them difficult to read.
- **[writing]** Figure 12: The y-axis label ‘MCC’ is missing from the individual subplots, relying on the user to infer the metric from the caption.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript is dense with domain-specific terminology and acronyms that are not consistently defined upon first use, creating a barrier for non-specialist readers or those from adjacent fields (e.g., general ML or biology). While the paper aims to be a comprehensive benchmark, the reliance on unexplained jargon undermines its accessibility.

Specific instances requiring attention include:

1. **Acronyms:** ‘MCC’ (Matthews Correlation Coefficient), ‘SSM’ (State Space Model), ‘BPE’ (Byte-Pair Encoding), ‘k-mer’, ‘MoE’ (Mixture of Experts), ‘T-enc’/‘T-dec’ (Transformer-encoder/decoder), ‘SN’ (Single-Nucleotide), ‘T2T’ (Telomere-to-Telomere), ‘eQTL’, ‘Hi-C’, ‘lncRNA’, ‘TF’ (Transcription Factor), ‘NT’ (Nucleotide Transformer), ‘GUE’ (Genome Understanding Evaluation), ‘GB’ (Genomic Benchmarks), ‘PGB’ (Plant Genomics Benchmark), ‘iPro’, ‘iDNA’, ‘iDHS’, ‘4mC’, ‘5mC’, ‘6mA’, ‘ReLU’, ‘CNN’ (Convolutional Neural Network), ‘Hyena’, ‘StripedHyena’, ‘Graph-Transformer’, ‘BioToken’. Many of these appear in tables and figure captions without prior definition in the main text.
2. **Domain Concepts:** ‘Pareto frontier’, ‘Spearman correlation’, ‘macro-MCC’ vs. ‘micro-MCC’, ‘logistic regression’, ‘few-shot’ (1-shot, 10-shot), ‘prokaryotic’, ‘eukaryotic’, ‘viral’, ‘phage’, ‘coding’, ‘non-coding’, ‘chromatin’, ‘enhancer’, ‘promoter’, ‘splice sites’, ‘histone modifications’, ‘DNA methylation’, ‘species classification’, ‘regulatory’, ‘TF binding’, ‘lncRNA’, ‘mouse enhancers’, ‘virus/phage’, ‘coding/non-coding’, ‘chromatin accessibility’. While some are standard in genomics, their specific usage in the context of these benchmarks (e.g., ‘mouse enhancers’ as a specific task category) should be clarified.
3. **Model Names:** ‘Mamba-SSM’, ‘HyenaDNA’, ‘GenomeOcean’, ‘Evo’, ‘DNABERT’, ‘GENA-LM’, ‘GROVER’, ‘Nucleotide Transformer’, ‘JanusDNA’, ‘Gene42’, ‘LucaOne’, ‘OmniNA’, ‘Enformer’, ‘DNALongBench’, ‘SPACE’, ‘BioToken’, ‘BioFM’, ‘DeepGene’, ‘Genomics-FM’, ‘PlantCaduceus’, ‘AgroNT’, ‘MutBERT’, ‘Evo2’, ‘ChatNT’, ‘NucleotideGPT’, ‘EpiGePT’, ‘ENBED’, ‘HAD’, ‘C.La.P.’, ‘HybridDNA’, ‘MxDNA’, ‘VQDNA’, ‘BMFM-DNA’. While these are proper nouns, the underlying architecture or data type (e.g., ‘Mamba-SSM’ implying a

State Space Model) should be briefly explained when first introduced.

The paper frequently uses abbreviations for task categories (e.g., ‘Histone Modifications’, ‘Promoters’, ‘Enhancers’, ‘DNA Methylation’, ‘Splice Sites’, ‘lncRNA’, ‘Mouse Enhancers’, ‘TF Binding’, ‘Species Classification’, ‘Regulatory’, ‘Virus/Phage’, ‘Coding/Non-coding’, ‘Chromatin Accessibility’) without ensuring the reader understands the specific biological or computational meaning in the context of the benchmark. For instance, ‘lncRNA’ is used extensively but not defined as ‘long non-coding RNA’ until perhaps much later or not at all in a clear, accessible way. Similarly, ‘TF binding’ is used without explicitly stating ‘Transcription Factor binding’ in all instances.

The use of ‘1-shot’, ‘10-shot’, and ‘full-shot’ is common in ML but should be explicitly defined as evaluation regimes with 1, 10, and all available labeled examples, respectively, for readers less familiar with few-shot learning terminology.

The term ‘Pareto frontier’ is used to describe the trade-off between model size and performance. While a standard concept in optimization, a brief explanation (e.g., ‘the set of models that are not outperformed by any other model in both size and performance’) would make the figure and its interpretation more accessible to a broader audience.

The distinction between ‘macro-MCC’ and ‘micro-MCC’ is crucial for understanding the benchmark’s aggregation strategy but is not clearly explained in plain language. A simple definition (e.g., ‘macro-MCC averages performance across categories, while micro-MCC averages across all individual tasks’) would be beneficial.

The paper also uses specific dataset names (e.g., ‘GUE’, ‘GB’, ‘NT’, ‘PGB’, ‘iPro’, ‘iDNA’, ‘iDHS’) as prefixes for tasks without always providing a clear, accessible definition of what these datasets represent or why they are grouped under these names.

In summary, the manuscript would significantly benefit from a more consistent and explicit definition of jargon, acronyms, and domain-specific concepts upon their first appearance. This would align with the paper’s goal of providing a standardized and accessible benchmark for the broader community.

Action items.

- **[writing]** Acronyms: ‘MCC’ (Matthews Correlation Coefficient), ‘SSM’ (State Space Model), ‘BPE’ (Byte-Pair Encoding), ‘k-mer’, ‘MoE’ (Mixture of Experts), ‘T-enc’/‘T-dec’ (Transformer-encoder/decoder), ‘SN’ (Single-Nucleotide), ‘T2T’ (Telomere-to-Telomere), ‘eQTL’, ‘Hi-C’, ‘lncRNA’, ‘TF’ (Transcription Factor), ‘NT’ (Nucleotide Transformer), ‘GUE’ (Genome Understanding Evaluation), ‘GB’ (Genomic Benchmarks), ‘PGB’ (Plant Genomics Benchmark), ‘iPro’, ‘iDNA’, ‘iDHS’, ‘4mC’, ‘5mC’, ‘6mA’, ‘ReLU’, ‘CNN’ (Convo
- **[writing]** Domain Concepts: ‘Pareto frontier’, ‘Spearman correlation’, ‘macro-MCC’ vs. ‘micro-MCC’, ‘logistic regression’, ‘few-shot’ (1-shot, 10-shot), ‘prokaryotic’, ‘eukaryotic’, ‘viral’, ‘phage’, ‘coding’, ‘non-coding’, ‘chromatin’, ‘enhancer’, ‘promoter’, ‘splice sites’, ‘histone modifications’, ‘DNA methylation’, ‘species classification’, ‘regulatory’, ‘TF binding’, ‘lncRNA’, ‘mouse enhancers’, ‘virus/phage’, ‘coding/non-coding’, ‘chromatin accessibility’. While some are standard in genomics, their s
- **[writing]** Model Names: ‘Mamba-SSM’, ‘HyenaDNA’, ‘GenomeOcean’, ‘Evo’, ‘DNABERT’,

‘GENA-LM’, ‘GROVER’, ‘Nucleotide Transformer’, ‘JanusDNA’, ‘Gene42’, ‘LucaOne’, ‘OmniNA’, ‘Enformer’, ‘DNALongBench’, ‘SPACE’, ‘BioToken’, ‘BioFM’, ‘DeepGene’, ‘Genomics-FM’, ‘PlantCaduceus’, ‘AgroNT’, ‘MutBERT’, ‘Evo2’, ‘ChatNT’, ‘NucleotideGPT’, ‘EpiGePT’, ‘ENBED’, ‘HAD’, ‘C.La.P.’, ‘HybriDNA’, ‘MxDNA’, ‘VQDNA’, ‘BMFM-DNA’. While these are proper nouns, the underlying architecture or data type (e.g., ‘Mamba-SSM’ implying a

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically coherent argument that aggregate benchmarking masks significant heterogeneity in genomic model performance. The central thesis—that scale is an imperfect predictor and that domain alignment (pretraining corpus) is critical—is well-supported by the provided statistical evidence (Spearman correlations, controlled pair differences). The distinction between macro-averaged and micro-averaged metrics is handled consistently, and the stability analyses (probe type, regularization) effectively rule out methodological artifacts as the primary driver of the observed rankings.

However, there are minor logical gaps in the derivation of specific counts and the strength of generalizations from limited controlled pairs. First, the claim of “31 instances” where smaller models outperform larger ones (Section 4) is a specific derived statistic that is not explicitly tabulated. While the correlation coefficient ($\rho = 0.565$) supports a general trend, the specific count of counter-examples requires a direct reference to a list or table to be fully verifiable. Second, the generalization that “Transformer-based models generally outperform state-space alternatives” (Conclusion) is primarily driven by a few high-profile comparisons (e.g., Omni-DNA vs. eccDNAMamba). The paper acknowledges that SSMs can be competitive (e.g., on Chromatin Accessibility), but the logical weight given to the “general” advantage could be better calibrated by explicitly stating the proportion of categories where Transformers won versus where SSMs were competitive. Finally, the claim that virus/phage tasks “favor human-only pretraining” (Section 4) is slightly overstated; the data shows a negative average difference, but only 2 out of 6 controlled pairs actually favored human-only models. The phrasing suggests a majority trend that the “Wins” column does not support, creating a minor inconsistency between the statistical summary and the textual conclusion. These issues are fixable by refining the text to more precisely reflect the underlying data distributions.

Action items.

- **[science]** The claim that ‘31 instances’ exist where a model 5x smaller outperforms a larger one (Section 4) lacks a direct citation to a table or figure listing these specific pairs. While the aggregate correlation is provided, the specific count of 31 is a derived statistic that requires explicit tabular evidence to verify the logical step from ‘correlation’ to ‘count of

counter-examples’.

- **[writing]** The conclusion that ‘Transformer-based models generally outperform the evaluated state-space alternative’ (Section 5) relies heavily on the comparison between Omni-DNA-1B and eccDNAMamba. However, the paper notes that eccDNAMamba outperforms GenomeOcean-500M on Chromatin Accessibility. The logical leap to a general ‘Transformer advantage’ without quantifying the frequency of SSM wins across the 13 categories weakens the causal claim.
- **[writing]** The statement that ‘Multi-species pretraining shows its most consistent advantage on chromatin accessibility (6/6 pairs)’ is supported, but the claim that ‘virus/phage tasks favor human-only pretraining’ is based on a negative average (-0.034) where only 2/6 pairs actually favored human-only. The phrasing ‘favor’ implies a majority trend which is not supported by the ‘Wins’ column (2/6), creating a potential logical inconsistency in the strength of the claim.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes a compelling case for the heterogeneity of genomic model performance, but several conclusions overreach the specific evidence provided by the benchmark’s scope and sample size.

First, the conclusion states that “Transformer-based models generally outperform the state-space alternative.” While the data supports this for the majority of categories and the specific controlled pairs analyzed (e.g., Omni-DNA vs. eccDNAMamba), the paper explicitly identifies a domain where SSMs are competitive or superior (Chromatin Accessibility, where eccDNAMamba outperforms GenomeOcean-500M). The use of “generally” without immediately qualifying the specific conditions under which SSMs succeed risks misleading readers into a binary architectural hierarchy that the data does not fully support. The claim should be tempered to emphasize that architectural superiority is highly task-dependent.

Second, the assertion that “domain mismatch cannot be overcome by scale” (Section: Transfer Learning Analysis) is an absolute negative claim. The evidence provided is the failure of a single 7B prokaryotic model (Evo-1-131k) on eukaryotic tasks. While this is a striking result, extrapolating from one data point to a universal law that scale *cannot* overcome domain mismatch is an overreach. It is possible that a larger prokaryotic model or a different training objective could yield different results. The text should be revised to state that “in the evaluated regime, scale did not compensate for domain mismatch” or “the observed data suggests scale alone is insufficient,” rather than stating it as an impossibility.

Third, the conclusion that “Microbial-only corpora transfer poorly to eukaryotic tasks” relies

heavily on the performance of DNABERT-S (microbial) compared to multi-species models and the prokaryotic Evo-1-131k. Given that the benchmark includes only one or two models with strictly microbial-only pretraining, presenting this as a definitive, generalizable principle without explicitly noting the limited sample size of such models in the study is a minor overreach. The authors should qualify this finding to reflect the specific constraints of their model coverage.

Finally, the claim that “The model that wins under full supervision reranks under 10-shot evaluation in 8 of 13 categories” (Conclusion) is a strong statistical claim. While the data likely supports this, the phrasing implies a systematic instability that might be better described as “significant rank instability” or “frequent reranking” to avoid implying a deterministic rule for 8/13 categories without a deeper statistical test of the stability of these specific rank changes.

These issues are primarily matters of phrasing and the strength of the logical leap from specific data points to general principles. The core science is sound, but the conclusions should be more precisely bounded by the specific limitations of the evaluated model set.

Action items.

- **[writing]** The claim that ‘Transformer-based models generally outperform the state-space alternative’ (Conclusion) overgeneralizes from a limited set of controlled pairs. While the paper notes exceptions (e.g., Chromatin Accessibility), the phrasing implies a universal architectural superiority not fully supported by the data, which shows SSMs can be competitive or superior in specific domains. Qualify this claim to reflect the domain-specific nature of the findings.
- **[writing]** The statement that ‘domain mismatch cannot be overcome by scale’ (Section: Transfer Learning Analysis) is an absolute claim that risks overreach. The data shows a catastrophic failure for one specific prokaryotic model (Evo-1-131k) on eukaryotic tasks, but this does not prove that *no* amount of scale could ever bridge such a gap. Rephrase to reflect that ‘in the evaluated regime, scale did not compensate for domain mismatch’ or similar.
- **[writing]** The conclusion that ‘Microbial-only corpora transfer poorly to eukaryotic tasks’ (Conclusion) is based on a single controlled pair (DNABERT-S vs. NT-v2) and the performance of Evo-1-131k. While the trend is clear, presenting this as a definitive principle without acknowledging the limited sample size of microbial-only models in the benchmark constitutes a slight overreach. Add a qualifier regarding the sample size.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses critical safety and ethical dimensions of genomic foundation models, particularly regarding the lack of standardized evaluation and the potential for biased or dual-use outcomes. The authors include a dedicated “Responsible Application and Ethical Considerations” section (e000) and an “Impact Statement” that correctly identify the risks of deploying these

models in clinical or ecological contexts without rigorous validation. The disclosure of no financial conflicts of interest is clear.

However, the ethical discussion remains somewhat generic. While the paper warns against deployment in clinical settings, it does not provide a concrete framework for how the GENE benchmark itself mitigates dual-use risks, such as the potential for these models to be repurposed for designing novel pathogens or toxins. Given the sensitivity of genomic data, a more robust “Dual-Use Risk Assessment” is required. This should explicitly address how the benchmark’s design (e.g., task selection, data sources) avoids facilitating harmful applications and what specific safeguards are recommended for users.

Furthermore, the “Impact Statement” notes that the task suite is skewed toward eukaryotic and human data. While this is a scientific limitation, the ethical implication—potential harm to underrepresented populations if these models are deployed in precision medicine without correction—needs to be articulated more forcefully. The authors should expand on the specific risks of bias propagation in clinical decision-making.

Finally, the funding source (Ministry of Economic Development of the Russian Federation) is disclosed, but given the global nature of genomic data and biosecurity concerns, the manuscript would benefit from a brief statement on compliance with international biosecurity standards and data sovereignty considerations. The current text is sufficient to flag these issues but requires expansion to meet the rigorous safety standards expected for a benchmark of this scale.

Action items.

- **[writing]** The ‘Responsible Application and Ethical Considerations’ section (e000) provides high-level warnings but lacks specific mitigation strategies for the identified dual-use risks (e.g., pathogen design). Explicitly detail how the benchmark prevents or detects misuse, or add a dedicated ‘Dual-Use Risk Assessment’ subsection with concrete safeguards.
- **[writing]** The ‘Impact Statement’ and ‘Limitations’ sections acknowledge dataset biases toward human and model organisms. However, the paper does not quantify the potential for these biases to cause harm in clinical or agricultural deployment (e.g., misdiagnosis in underrepresented populations). Add a specific analysis of the downstream risks associated with these documented biases.
- **[writing]** The ‘Acknowledgements’ section lists funding from the Ministry of Economic Development of the Russian Federation. While a conflict of interest is declared as ‘none’, the geopolitical context of genomic research requires a more explicit statement regarding data sovereignty, potential export control implications, and adherence to international biosecurity norms.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a large-scale benchmark (GENEB) with a substantial sample size (40 models, 100 tasks), which is a significant strength. The use of controlled pair comparisons (30 pairs) to isolate variables like architecture and pretraining data is a robust methodological choice that strengthens the causal claims regarding transfer learning and architectural differences. The statistical reporting (Spearman correlations, p-values) is generally appropriate for the scale of the data.

However, the strength of the evidence is compromised in specific areas where the sample size for the specific claim is too small to support the generalization made. Most notably, the conclusion regarding the superiority of “eukaryotic-gene focused” pretraining is based on a single controlled pair (Appendix, Table 4). While the effect size is large, $n=1$ is insufficient to establish a general principle, especially given the heterogeneity of the other models. Similarly, the specific claim of “31 instances” of smaller models outperforming larger ones (Section 4) is presented without a supporting list or table, making it impossible to verify if this count is accurate or if it relies on a specific, potentially biased, definition of “outperform.”

Furthermore, the observation of an “inverse performance pattern” in few-shot robustness (where lower-performing models show less degradation) requires a more rigorous check against floor effects. Since many tasks (e.g., DNA methylation, lncRNA) result in near-random performance for all models in the 1-shot regime, the apparent stability of the worst models may simply be a mathematical artifact of having no room to drop further, rather than a genuine property of their representations. The current text does not adequately address this alternative explanation.

Finally, while the paper correctly identifies that the prokaryotic-only model (Evo-1-131k) is an outlier, the exclusion of this model to boost the correlation coefficient from 0.565 to 0.685 should be accompanied by a sensitivity analysis showing how the ranking of the top models changes with and without this outlier, to ensure the “scale-performance” conclusion is not driven entirely by the removal of a single domain-mismatched data point.

Action items.

- **[science]** The claim that ‘31 instances’ exist where a model 5x smaller outperforms a larger one (Section 4, ‘The Scale-Performance Disconnect’) lacks a supporting table or list. Without a concrete enumeration of these pairs, the specific count is unverifiable and risks being an artifact of cherry-picking. Please provide a supplementary table listing these 31 pairs with their respective sizes and MCC scores.
- **[science]** The controlled pair analysis for ‘Eukaryotic-genes vs. multi-species’ relies on a single pair (GENERator-Eukaryote-3B vs. DNA-GPT-3B-M) (Appendix, Table 4). Drawing a general conclusion about the superiority of gene-focused pretraining from $n=1$ is statistically weak. Please either expand this analysis to include more matched pairs or significantly temper the language to reflect that this is a single-case observation rather than a robust trend.
- **[science]** The few-shot robustness analysis notes an ‘inverse performance pattern’ where smaller/worse models degrade less (Section 4). However, the paper does not explicitly test whether this is a statistical artifact of the floor effect (models already near random performance cannot drop much further). Please include a control analysis or discussion ruling

out floor effects as the primary driver of this observation.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis in GENEb is generally rigorous, employing appropriate metrics (MCC) and robust evaluation protocols (5 seeds, logistic regression). The use of Spearman rank correlation to assess scaling laws is suitable for the non-linear relationship between parameter count and performance. However, the manuscript relies heavily on point estimates without providing measures of uncertainty, which weakens the statistical support for several key claims.

First, the correlation analyses in Table 1 (Per-category scaling correlations) report p-values but omit confidence intervals for the Spearman ρ coefficients. With a sample size of $n = 40$, the confidence intervals for correlations near the significance threshold (e.g., Species Classification, $p = 0.057$) are likely wide. Reporting 95% CIs would clarify whether the lack of significance is due to low power or a genuinely weak relationship.

Second, the “Controlled-Pair Comparisons” (Appendix, Section 30 pairs) aggregate results from very small subsets (e.g., $n = 6$ pairs for human vs. multi-species). While the mean Δ MCC is reported, the absence of standard errors or confidence intervals makes it difficult to assess the reliability of these differences. For instance, the claim that multi-species pretraining yields a $+0.012$ aggregate advantage relies on averaging only 6 pairs; a single outlier pair could significantly alter this conclusion.

Third, the “inverse performance pattern” in few-shot robustness (Section 4, paragraph 4) asserts that weaker models degrade less. This is a correlation claim (performance vs. degradation) that requires a statistical test (e.g., Pearson/Spearman correlation with p-value and CI) rather than just a descriptive list of the top 5 models. Without this, the pattern remains anecdotal.

Finally, the probe stability analysis (Appendix) correctly uses rank correlation to show robustness, but the absolute MCC shifts (mean $+0.011$) should be accompanied by a distribution (e.g., boxplot or range) to ensure the “small” shifts are not masking large variances in specific tasks. Overall, adding confidence intervals and formal significance tests for the comparative claims would significantly strengthen the statistical validity of the paper.

Action items.

- **[science]** The paper reports Spearman correlations for scaling laws (e.g., Table 1) but omits confidence intervals. Given the $n=40$ sample size, CIs are essential to assess the precision of these estimates and the robustness of the ‘significant’ claims, particularly for categories with p-values near 0.05 (e.g., Species Classification $p=0.057$).
- **[science]** The controlled pair comparisons ($n=6$ for human vs. multi-species) rely on small sample sizes to draw broad conclusions about pretraining corpus effects. The manuscript

should report the variance (e.g., standard error or 95% CI) of the mean delta-MCC to demonstrate that the observed effects are not driven by outliers or high variance within the pairs.

- **[science]** The analysis of few-shot degradation (Figure 3) presents mean MCC drops but lacks statistical testing for the ‘inverse performance pattern’ claim. A formal test (e.g., correlation between full-shot performance and degradation magnitude) with a p-value and confidence interval is required to support the assertion that smaller drops occur in weaker models.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical rigor and presents a comprehensive benchmark with clear, data-driven insights. The writing is generally precise, professional, and well-structured, effectively conveying complex comparative analyses of genomic models. The logical flow from the introduction of the problem (fragmented comparisons) to the methodology, results, and specific recommendations is coherent and easy to follow.

However, there are minor stylistic inconsistencies and areas where sentence structure could be tightened to improve readability for a broader audience. Specifically, the use of mathematical inequality symbols ($>$, $<$) within running prose in the “Results Analysis” appendix section disrupts the narrative flow and should be converted to standard English phrasing. Additionally, the “Practitioner Recommendations” section suffers from inconsistent punctuation in its list format, which detracts from the professional polish of the document.

There are also a few instances of slightly ambiguous phrasing, particularly in the “Few-Shot Robustness” section where the grouping of models (“and others”) could be clarified to ensure the reader immediately understands which specific models are being referenced. Finally, some redundancy exists between figure captions and the main text, particularly regarding the interpretation of high-variance tasks, which could be streamlined. Addressing these minor points will enhance the overall clarity and readability of the paper without altering its scientific content.

Action items.

- **[writing]** In Section ‘Results Analysis’ (Appendix), the sentence ‘Taxonomic alignment drives performance: multi-species $>$ human-only microbial $>$ prokaryotic’ uses mathematical inequality symbols in running prose. Replace with ‘multi-species outperforms human-only, which is comparable to microbial, which in turn outperforms prokaryotic’ for better readability.
- **[writing]** In the ‘Practitioner Recommendations’ section, the list of categories uses inconsistent punctuation. Some items end with periods while others do not (e.g., ‘Coding/non-coding’ vs ‘Promoter prediction’). Standardize to end every item with a period for consistency.

- **[writing]** In the ‘Few-Shot Robustness’ paragraph, the phrase ‘The five smallest absolute drops occur in Evo-1-131k... and others, all among the weakest models in full-shot’ is slightly ambiguous. Clarify that ‘others’ refers to the specific models listed immediately prior to avoid confusion about which models are being grouped.
- **[writing]** In the ‘Transfer Learning Analysis’ subsection, the sentence ‘The result indicates that high-variance tasks expose biologically meaningful differences...’ appears in the caption of Figure 5 but is also repeated in the main text. Ensure the main text refers to the figure rather than restating the caption verbatim to avoid redundancy.